

Project Overview & Context: The Transparency Gap in Technical Hiring



Resumes



PDF Uploads



LinkedIn Profiles

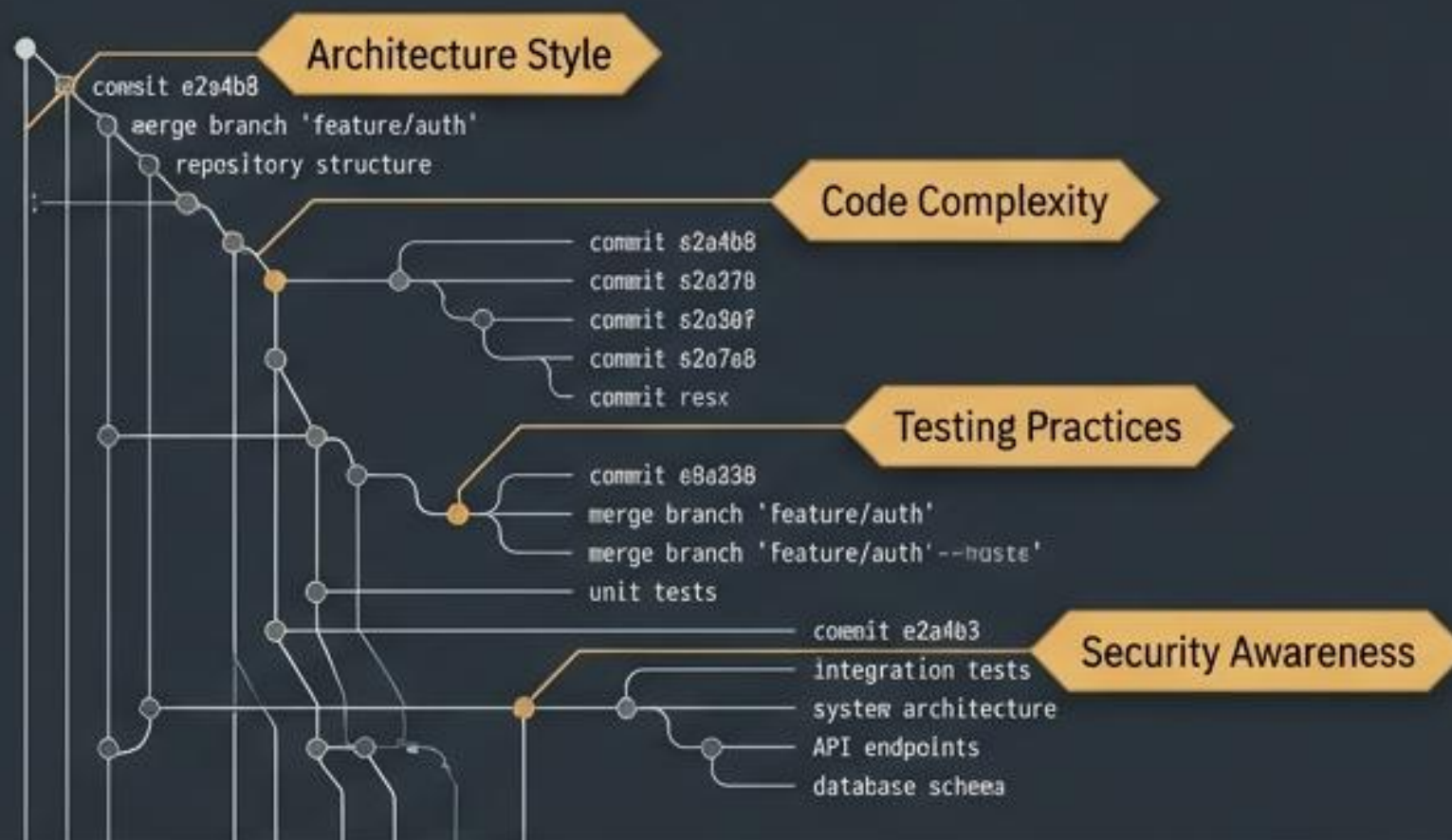
Surface Level

Stated Experience

Easily spoofed, highly keyword-optimized, and generating critical signal-to-noise ratios.

Current systems scrape only the surface, relying on manual screening and opaque technical assessments prone to pedigree bias.

Subsurface Reality

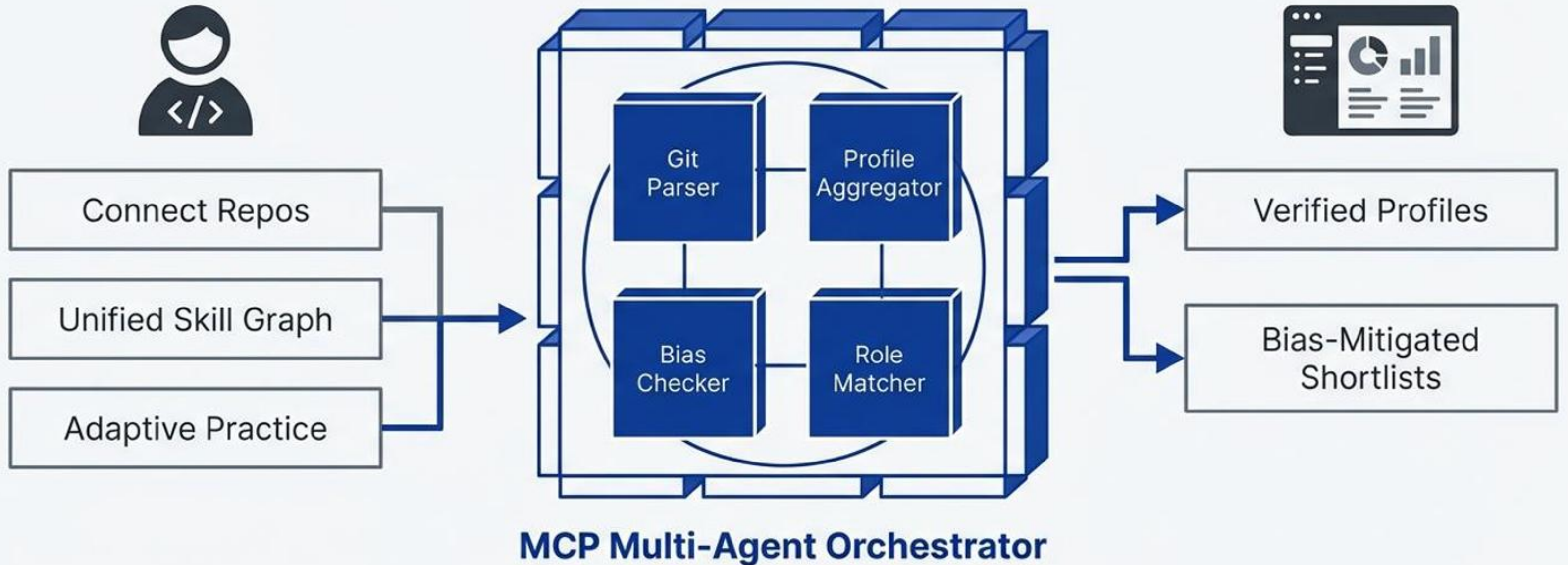


Demonstrated Experience

The hidden reality of candidate capability.

AegisHire dives into the repository to measure actual engineering reality, cross-referencing stated claims against demonstrated codebase commits to expose the critical resume-versus-repo gap.

A dual-sided ecosystem powered by verifiable intelligence



100% Explainable AI. Zero Black Boxes. Every score includes a human-readable explanation.

PROJECT MANAGEMENT & METHODOLOGY

Pragmatic Rollout Strategy & Agile Execution

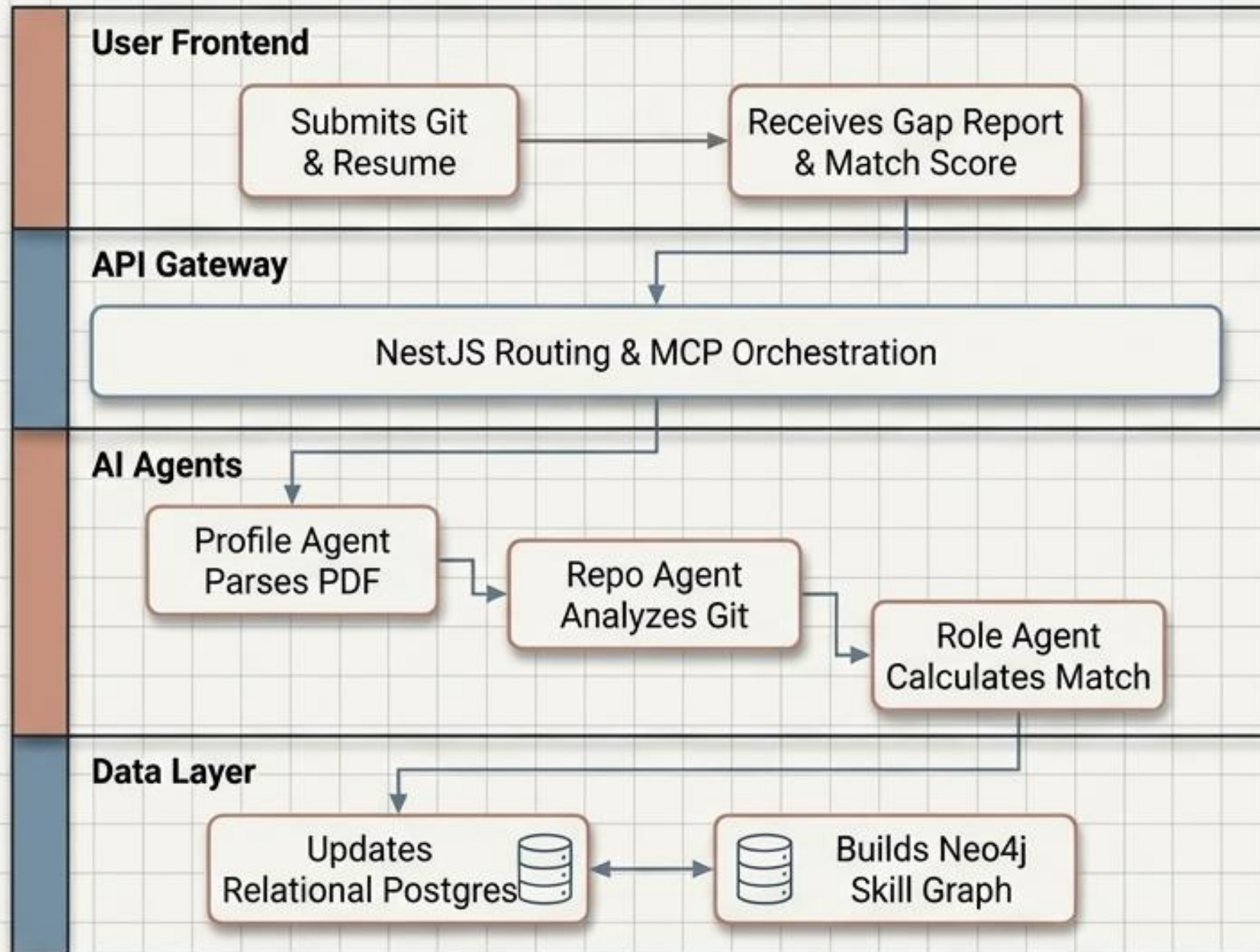


THE V1 VS V2 SCOPE MATRIX

Version 1: Establishing the Core Baseline	Version 2: Scaling Intelligence & Enterprise Live Edge
Repository Analysis & Profile Aggregation Agents	Live Code Editor with Anomaly Scoring
Neo4j Developer Skill Graph Generation	Voice & Behavioral Monitoring (Whisper AI)
Role Matching Engine & Gap Report System	Advanced Statistical Bias Auditing
Candidate Self-Assessment Dashboard	Live LinkedIn Integration Analytics
Focus: High structural value, low risk profile.	Focus: High-complexity enterprise features and consent-gated live monitoring.

REQUIREMENTS & DESIGN

System Architecture & Evaluation Lifecycle



GDPR Compliance

Data minimization, candidate data ownership, and strict opt-in consent flows.

AI Explainability

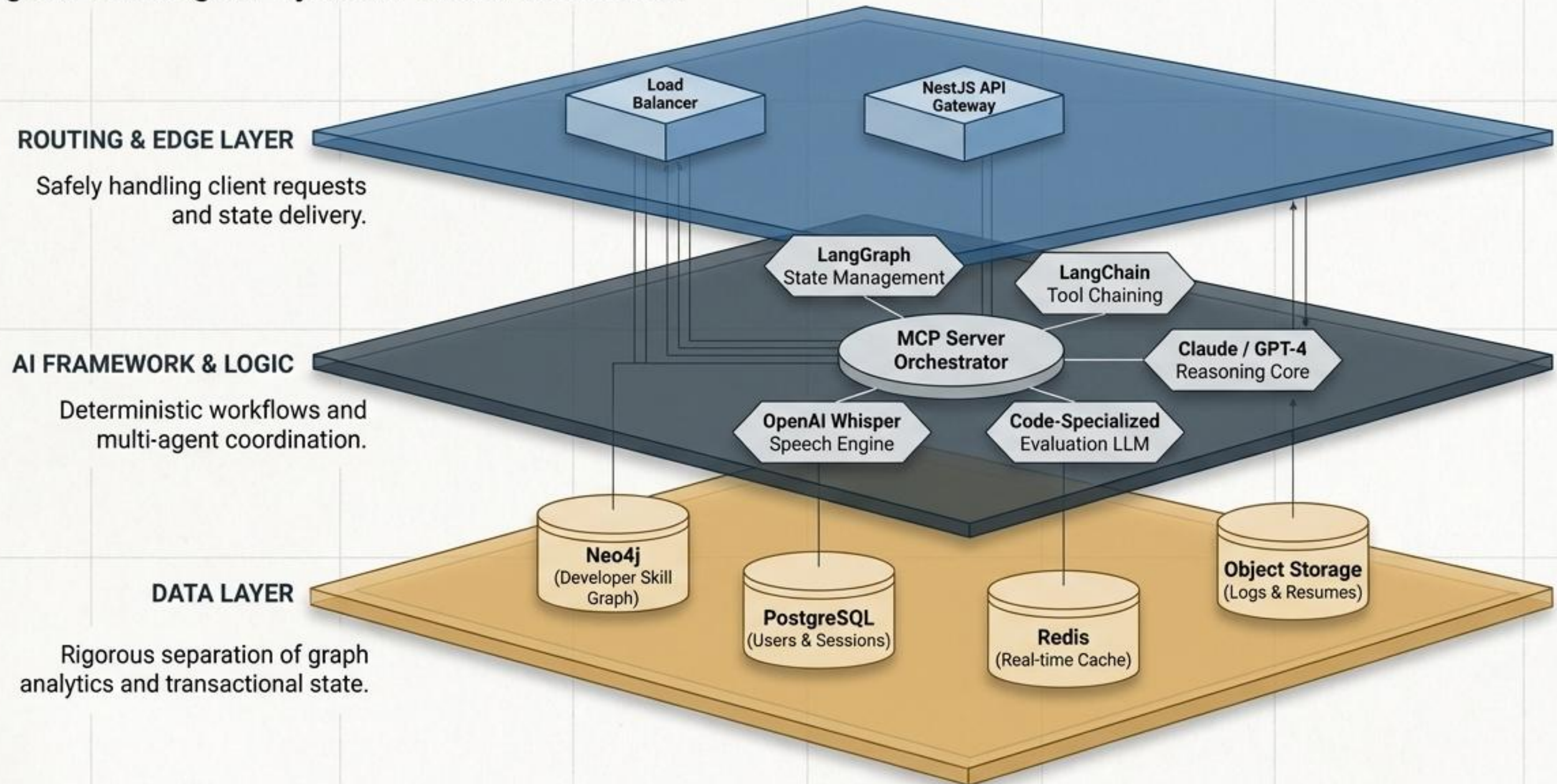
Probabilistic scoring only. Every AI decision surfaces a human-readable reason.

Scalability & Cost

Tiered LLM routing, token usage tracking, and aggressive graph query caching.

SYSTEM ARCHITECTURE

Logical Routing & Physical Persistence Stack



DEVELOPMENT ENVIRONMENT

Hardware & Software Assembly Stack

CORE BACKEND & ROUTING

NestJS

TypeScript-based enterprise API framework providing robust microservices architecture.

Safety Pipeline

Custom LLM Guardrails layer ensuring deterministic and probabilistic output validation.

AI & ORCHESTRATION LOGIC

Python Environment

Primary runtime for intelligence processing and agent logic.

LangGraph & LangChain

Deterministic state management and advanced LLM tool-chaining.

Haystack & LlamaIndex

Specialized retrieval pipelines for complex document parsing.

LLMs & INTELLIGENCE MODELS

Claude 3.5 Sonnet / GPT-4o

Heavyweight reasoning models for gap analysis and logic.

Code-Specialized LLMs

Lightweight models tailored exclusively for repository evaluation.

OpenAI Whisper

High-fidelity asynchronous speech-to-text processing.

DATABASE ECOSYSTEM

Neo4j

Native graph database mapping the complex nodes of the Developer Skill Graph.

PostgreSQL

Highly structured relational storage for transactional user profiles.

Redis

In-memory caching layer ensuring ultra-low latency for live interviews.